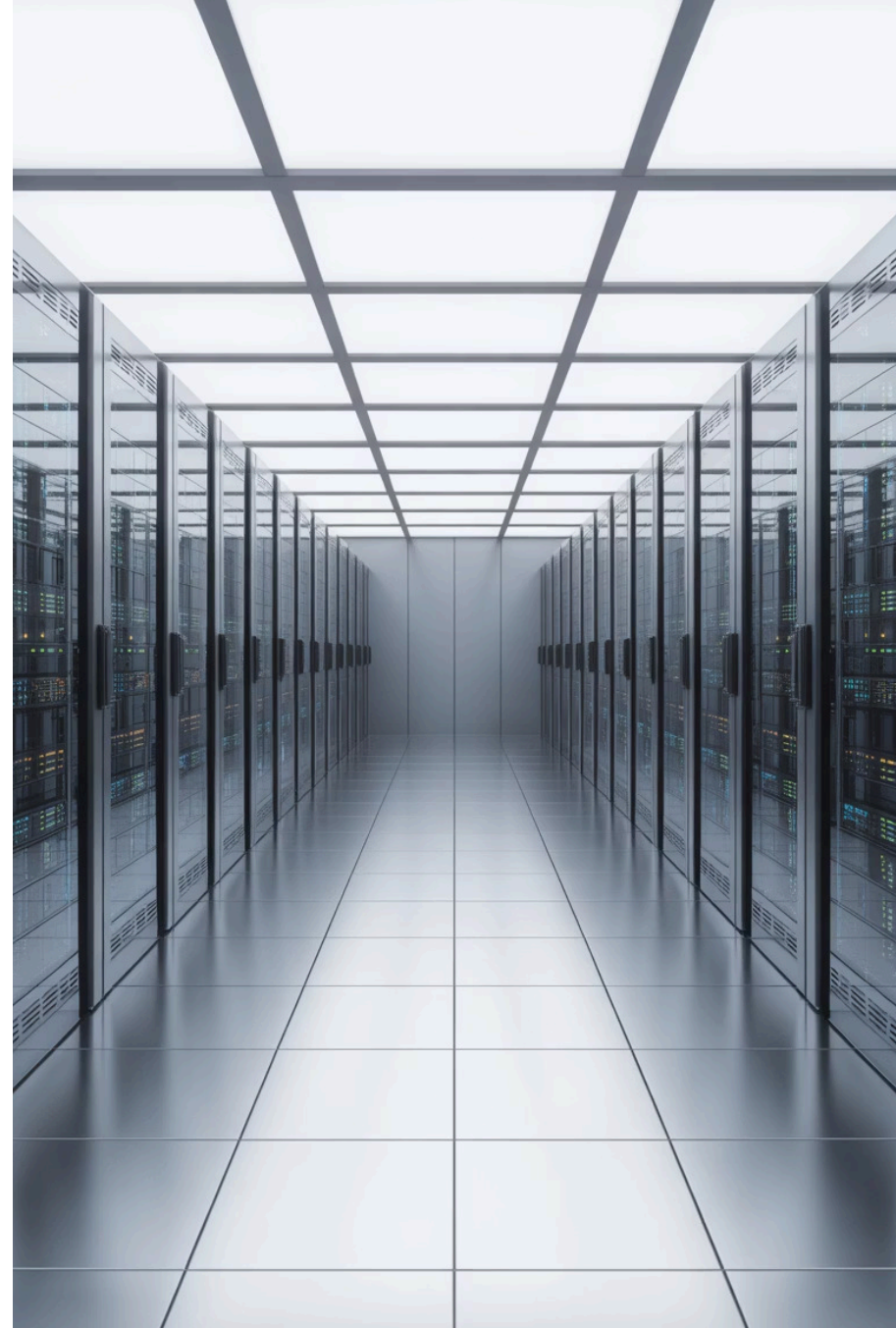


Google Agent Development Kit (ADK)

From Prototype to Production Agents



What is ADK?

Production-Ready Framework

Purpose-built for deploying real AI agents at scale, not just prototypes

Model-Agnostic Design

Bring your own models and providers without vendor lock-in

Modern Developer Experience

Feels like building standard software with familiar tooling and patterns



Enterprise-Ready from Day One

Official Language Support

- **Python** – Fast iteration and ML ecosystem integration
- **Java** – Enterprise backends and existing service integration
- Strong ecosystem with long-term maintainability guarantees

Why It Matters

Standardizes orchestration, tools, and evaluation across your organization. Fits seamlessly into enterprise guardrails and existing infrastructure without requiring architectural changes.





Deploy Anywhere Containers Run



Local Development

Fast iteration on your machine with full tooling



Cloud Run & VMs

Seamless deployment to managed services



Kubernetes

Enterprise orchestration at any scale



Vertex AI Integration

Compatible with Agent Engine or your own stack

Container-first architecture enables CI/CD-friendly, repeatable builds across any cloud provider or on-premises infrastructure.

Developer Experience & Observability

Consistent Project Structure

Familiar patterns and tooling reduce onboarding time and cognitive load

Built-In Tracing & Metrics

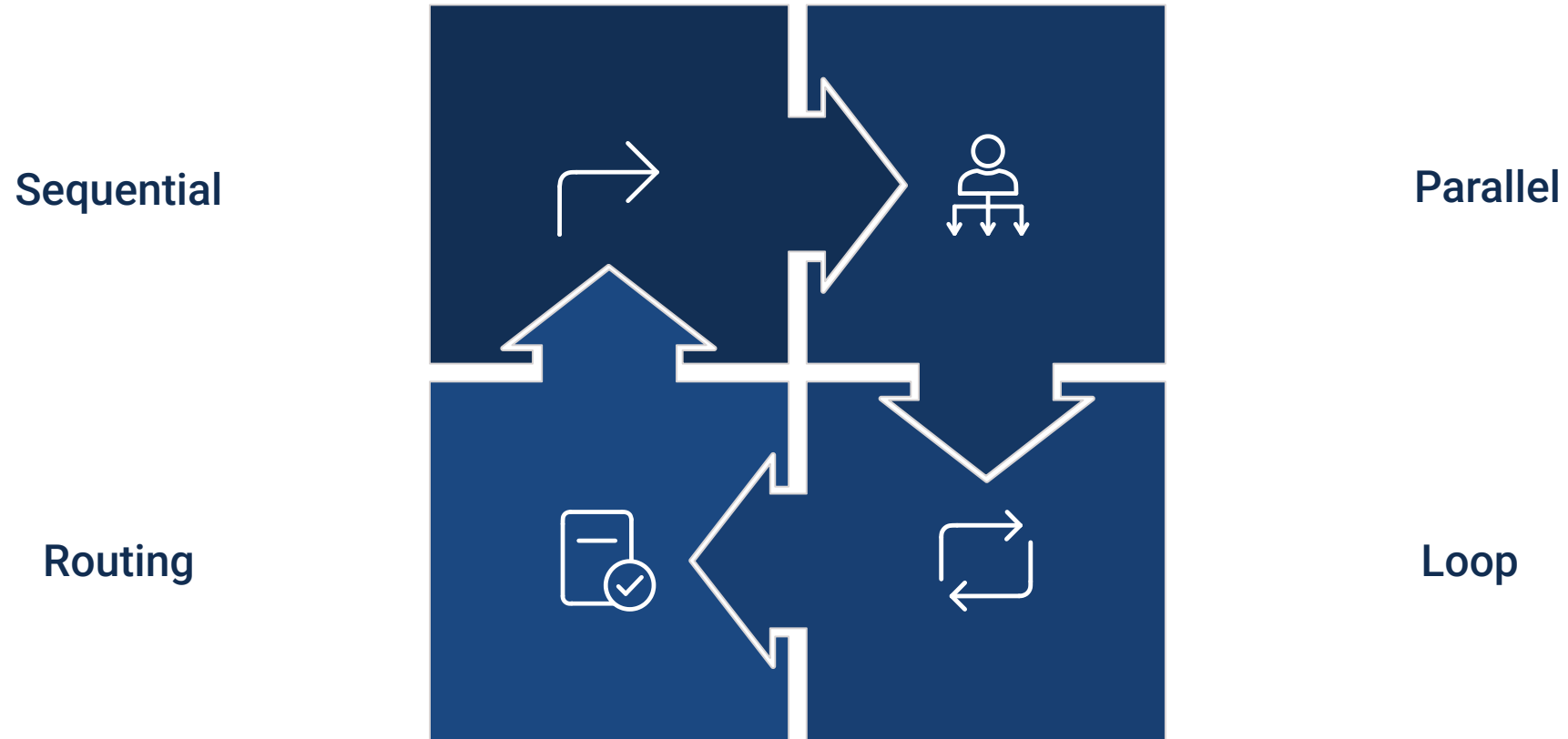
Logs, traces, and metrics ship out of the box—no custom instrumentation needed

Fast Iteration Cycles

Clear failure modes and debugging tools reduce time-to-fix more than any model tweak

📌 **DX Matters:** Great observability reduces debugging time by 10x compared to bespoke glue code that's hard to audit and scale.

Orchestration That Scales



Core Primitives

- **Sequential** – Step-by-step execution
- **Parallel** – Concurrent task processing
- **Loops** – Iterative workflows
- **Dynamic Routing** – LLM-driven decisions

Multi-Agent Systems

Compose agent hierarchies and teams. As scope grows, orchestration primitives keep complexity legible even as agents multiply across your system.



Rich Tools & Integrations

01

Prebuilt Tools

Search, code execution, and common integrations ready to use

02

Custom Functions

Write your own tools with simple function decorators

03

Agents as Tools

Compose specialized agents into larger systems

04

HTTP Tool Servers

Clean separation with language-agnostic service boundaries

Evaluation & Safety Built In



Final Response Testing

Validate end-to-end answers against expected outputs and quality criteria



Trajectory Analysis

Inspect step-by-step reasoning paths to catch errors early in the chain



Safety Patterns

Secure defaults and best practices baked into the framework architecture

Shift-left on quality: Evaluation keeps you honest across versions and datasets, catching regressions before production.



Interoperability & Flexibility



Framework Compatible

Works alongside other agent frameworks without forcing a rewrite



Model Freedom

Bring your own models and providers—OpenAI, Anthropic, Gemini, or custom



Minimal Lock-In

Designed by principle to avoid proprietary dependencies and vendor constraints

ADK gives you production controls, observability, and portability without sacrificing flexibility. Avoid bespoke glue code that becomes impossible to scale and audit.

Demo: Data Agent

- 1 Natural Language Query**
User asks a business question in plain English
- 2 Tool Execution**
Agent calls SQLite via HTTP tool server with real queries
- 3 ADK Web UI**
Visualize the full execution flow from question to answer

[Check the demo on github](#)

 **Ready to Start?** Begin locally with containers, add observability early, and evolve to multi-agent workflows as your scope grows. ADK scales with you from prototype to production.

[Explore ADK on GitHub](#)

[View Documentation](#)

